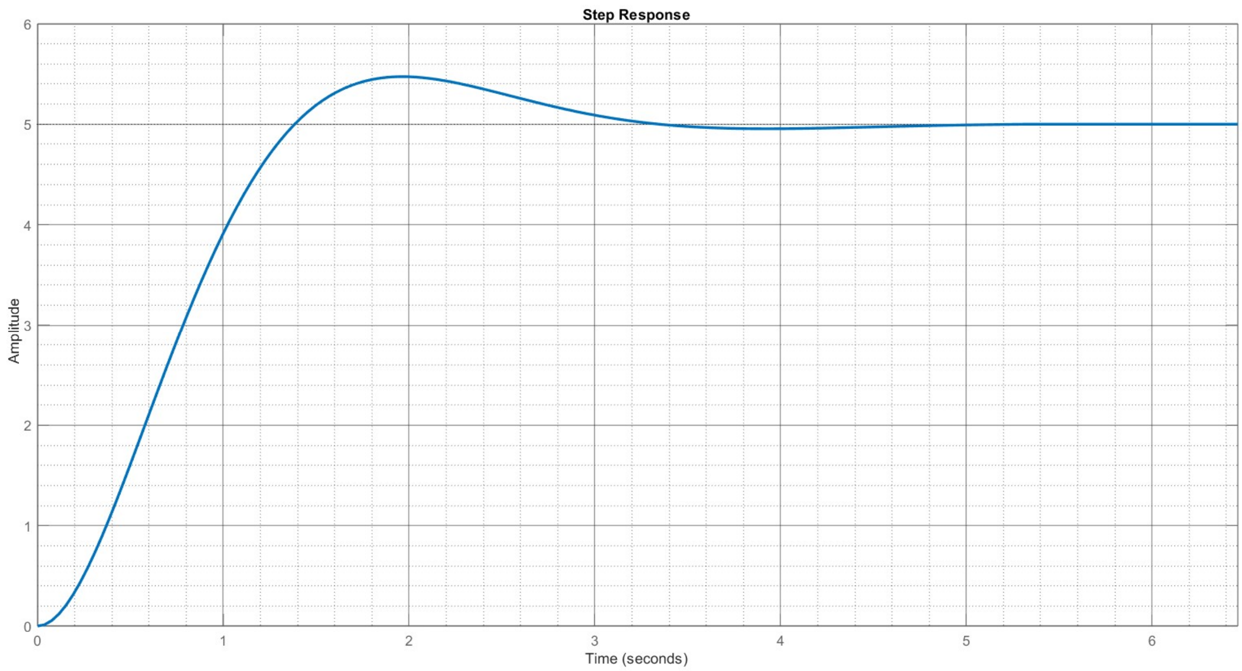
Name: ______________________________

Student no: ______________________________

Tut Test 2

The step response of a system is shown below. Based on your evaluation of the system's components, you think that it can be modelled as either a first-order system $G(s) = \frac{A}{\tau s+1}$ or a second-order system $G(s) = \frac{A\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$.

a) Explain which model you will be using, with a reason (2 marks), and hence
b) Estimate the unknown values in the model based on the step response. (8 marks)



You may use the formulas below to help you:

$$\frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \qquad \omega_d = \omega_n\sqrt{1-\zeta^2} \qquad \cos\theta = \zeta$$

$$t_p = \frac{\pi}{\omega_d} \qquad y_p = A\left(1 + e^{-\zeta\pi/\sqrt{1-\zeta^2}}\right) \qquad t_r = \frac{\pi - \theta}{\omega_d}$$

$$t_{k\%} = \frac{\ln\left(\frac{k}{100}\sqrt{1-\zeta^2}\right)}{-\zeta\omega_n} \qquad \%OS = 100e^{\left(\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}\right)}$$

Solution:

a) The system is at least second-order as the response exhibits overshoot/oscillation.
b) The gain $A = 5$, as this is the final value of the step response (2 marks).

The time to peak for the response is around 2 seconds, so $\frac{\pi}{\omega_d} = 2$, giving $\omega_d = \frac{\pi}{2}$. (2 marks)

The peak value is around 5.5. The formula for y_p or for the percentage overshoot can be used to determine ζ from this. Using the peak value formula:

$$5\left(1 + e^{-\zeta\pi \big/ \sqrt{1-\zeta^2}}\right) = 5.5 \therefore 1 + e^{-\zeta\pi \big/ \sqrt{1-\zeta^2}} = 1.1 \therefore e^{-\zeta\pi \big/ \sqrt{1-\zeta^2}} = 0.1$$

Taking the natural log of both sides:

$$-\zeta\pi \Big/ \sqrt{1-\zeta^2} = \ln(0.1) \therefore \zeta^2\pi^2 = (1-\zeta^2)\ln^2(0.1) \therefore \zeta^2(\pi^2 + \ln^2(0.1)) = \ln^2(0.1)$$

$$\therefore \zeta = \sqrt{\frac{\ln^2(0.1)}{\pi^2 + \mathrm{l}\ \ ^2(0.1)}} = 0.59 \text{ (3 marks)}$$

This gives $\omega_n = \frac{\omega_d}{\sqrt{1-\zeta^2}} = \frac{0.5\pi}{\sqrt{1-0.59^2}} = 1.95$ (1 mark)

(The actual values selected were $\zeta = 0.6$ and $\omega_n = 2$, so any good approximation that is supported by readings from the step response is fine.)